

- Projet Réseau – WAN Simulé
 - Documentation Technique
 - Guide de configuration réseau – WAN Simulé
 - 1. Objectif du projet
 - 2. Schéma du réseau
 - 3. Configuration des équipements Cisco
 - 3.1 Attribution des adresses IP des serveurs
 - 3.2 Configuration des switches
 - 3.3 Configuration des routers
 - 4. Test de réussite
 - 4.1 Relay et attributions IP des serveurs DHCP
 - 4.2 Test et configuration HTTP/DNS
 - 4.3 Test TFTP
 - 4.4 Test SYSLOG et Standby
 - 4.5 Test PING entre VLAN

Projet Réseau – WAN Simulé

| Documentation Technique

Auteur : Garlens Charles

Formation : BTS SIO – Option SISR

Année : 2025

Guide de configuration réseau – WAN Simulé

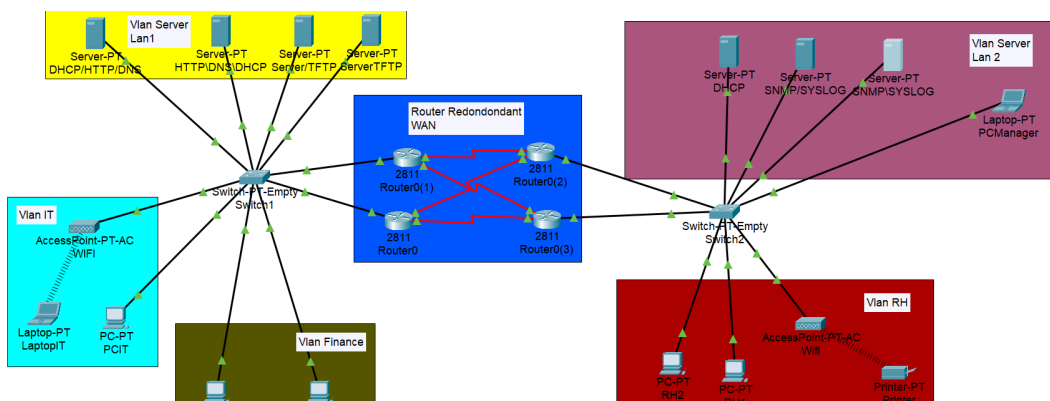
1. Objectif du projet

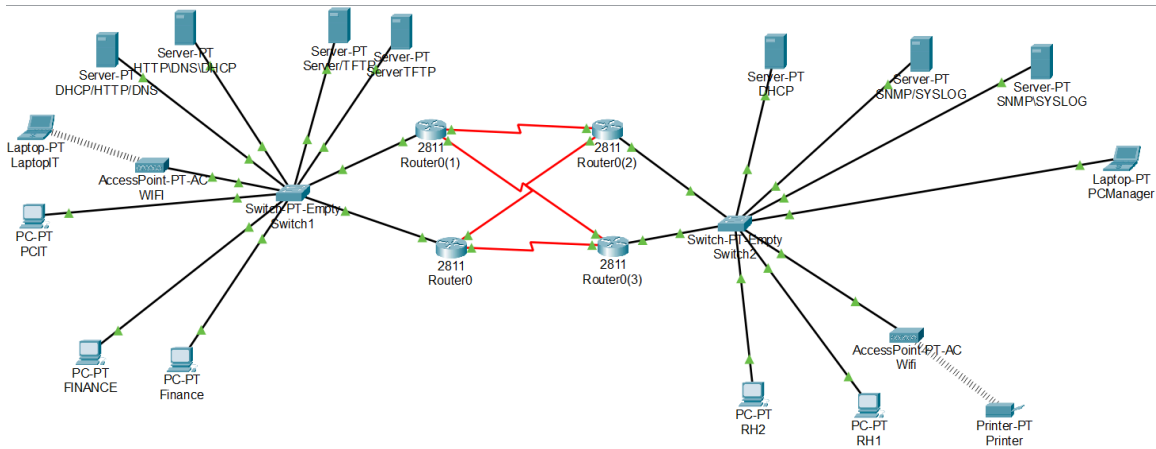
La PME **NeoEdu** possède deux agences distantes reliées par une liaison WAN simulée. Elle souhaite mettre en place :

- un réseau hautement disponible
- une redondance complète
- des VLANs par département
- une automatisation des sauvegardes
- une supervision avancée (SNMP, Syslog)

L'objectif est d'assurer **sécurité, performance, diagnostic automatisé**, et de produire une documentation complète.

2. Schéma du réseau





3. Configuration des équipements Cisco

3.1 Attribution des adresses IP des serveurs

```
IP Server DHCP = 192.168.80.4 255.255.255.0
IP Server SNMP/SYSLOG = 192.168.80.5 255.255.255.0
IP Server SNMP\SYSLOG = 192.168.80.6 255.255.255.0
IP Server DHCP/HTTP/DNS = 192.168.90.4 255.255.255.0
IP Server DHCP\HTTP/DNS = 192.168.90.5 255.255.255.0
IP ServerTFTP = 192.168.90.7 255.255.255.0
IP Server/TFTP = 192.168.90.6 255.255.255.0
```

3.2 Configuration des switches

```
# Switch 1

vlan 3
name Finance
```

```
vlan 4
name IT
vlan 90
name SERVER

interface g0/1
    switchport mode trunk
    switchport trunk allowed vlan 3,4,90
    no shutdown
    exit

interface g2/1
    switchport mode access
    switchport access vlan 4
    no shutdown
    exit

interface g3/1
    switchport mode access
    switchport access vlan 90
    no shutdown
    exit

interface g4/1
    switchport mode access
    switchport access vlan 90
    no shutdown
    exit

interface g5/1
    switchport mode access
    switchport access vlan 90
    no shutdown
    exit
```

```
interface g6/1
  switchport mode access
  switchport access vlan 90
  no shutdown
  exit
```

```
interface g7/1
  switchport mode access
  switchport access vlan 4
  no shutdown
  exit
```

```
interface g8/1
  switchport mode access
  switchport access vlan 3
  no shutdown
  exit
```

```
interface g9/1
  switchport mode access
  switchport access vlan 3
  no shutdown
  exit
```

```
write memory
```

```
# Switch 2
```

```
vlan 2
  name RH
vlan 80
  name Server
```

```
interface g0/1
```

```
switchport mode trunk
switchport trunk allowed vlan 2,80
no shutdown
exit
```

```
interface g3/1
switchport mode access
switchport access vlan 2
no shutdown
exit
```

```
interface g4/1
switchport mode access
switchport access vlan 2
no shutdown
exit
```

```
interface g9/1
switchport mode access
switchport access vlan 2
no shutdown
exit
```

```
interface g5/1
switchport mode access
switchport access vlan 80
no shutdown
exit
```

```
interface g6/1
switchport mode access
switchport access vlan 80
no shutdown
exit
```

```

interface g7/1
  switchport mode access
  switchport access vlan 80
  no shutdown
  exit

interface g8/1
  switchport mode access
  switchport access vlan 80
  no shutdown
  exit
write memory

```

Device Name: Switch1
 Device Model: Switch-PT-Empty
 Hostname: Switch

Port	Link	VLAN	IP Address	MAC Address
GigabitEthernet0/1	Up	--	--	0001.9633.AEC0
GigabitEthernet1/1	Up	--	--	0090.2B7B.51DE
GigabitEthernet2/1	Up	4	--	00E0.A3EA.45D5
GigabitEthernet3/1	Up	90	--	00D0.BA09.D8A9
GigabitEthernet4/1	Up	90	--	0001.C9D2.D4B0
GigabitEthernet5/1	Up	90	--	0002.170D.E8E5
GigabitEthernet6/1	Up	90	--	0090.21BB.6E06
GigabitEthernet7/1	Up	4	--	00D0.BC8D.E49B
GigabitEthernet8/1	Up	3	--	0050.0FBB.6861
GigabitEthernet9/1	Up	3	--	00E0.F9BC.3952
Vlan1	Down	1	<not set>	0007.EC96.6524

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Switch1

Device Name: Switch2
 Device Model: Switch-PT-Empty
 Hostname: Switch

Port	Link	VLAN	IP Address	MAC Address
GigabitEthernet0/1	Up	--	--	0090.0CC6.E62B
GigabitEthernet1/1	Down	1	--	0007.EC96.645A
GigabitEthernet2/1	Up	--	--	00D0.BC98.99A8
GigabitEthernet3/1	Up	2	--	00E0.B0CC.84E3
GigabitEthernet4/1	Up	2	--	00D0.97E2.B6C2
GigabitEthernet5/1	Up	80	--	00D0.D378.C684
GigabitEthernet6/1	Up	80	--	0000.0C37.98B9
GigabitEthernet7/1	Up	80	--	0006.2A65.A939
GigabitEthernet8/1	Up	80	--	000A.F32A.5D5C
GigabitEthernet9/1	Up	2	--	0090.0C5E.28BE
Vlan1	Down	1	<not set>	0005.5E9D.5792

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Switch2

3.3 Configuration des routers

```
# Router 1
```

```
enable
```

```
configure terminal
```

```
interface fa0/0
```

```
ip address 192.168.1.2 255.255.255.0
```

```
no shutdown
```

```
exit
```

```
# Router on a stick - Vlan Finance
```

```
interface fa0/0.3
```

```
encapsulation dot1Q 3
```

```
ip address 192.168.3.2 255.255.255.0
```

```
ip helper-address 192.168.90.5
```

```
ip helper-address 192.168.90.4
```

```
no shutdown
```

```
exit
```

```
# Vlan IT
```

```
interface fa0/0.4
```

```
encapsulation dot1Q 4
```

```
ip address 192.168.4.2 255.255.255.0
```

```
ip helper-address 192.168.90.5
```

```
ip helper-address 192.168.90.4
```

```
no shutdown
```

```
exit
```



```
# Vlan SERVER
```

```
interface fa0/0.90
  encapsulation dot1Q 90
  ip address 192.168.90.2 255.255.255.0
  no shutdown
  exit
```

```
# Redondance Router 1 Actif
```

```
interface fa0/0
  standby 1 ip 192.168.1.1
  standby 1 priority 110
  standby 1 preempt
  no shutdown
  exit
```

```
interface fa0/0.3
  standby 1 ip 192.168.3.1
  standby 1 priority 110
  standby 1 preempt
  no shutdown
  exit
```

```
interface fa0/0.4
  standby 1 ip 192.168.4.1
  standby 1 priority 110
  standby 1 preempt
  no shutdown
  exit
```

```
interface fa0/0.90
  standby 1 ip 192.168.90.1
  standby 1 priority 110
```

```
standby 1 preempt
no shutdown
exit
```

4. Test de réussite

4.1 Relay et attributions IP des serveurs DHCP

Avant arrêt d'un des serveurs :

☒ DHCP	☐ Static	DHCP request successful.
IPv4 Address	192.168.3.6	
Subnet Mask	255.255.255.0	
Default Gateway	192.168.3.1	
DNS Server	192.168.90.4	

PCVLAN3

Après la reprise totale des attributions par l'autre serveur :

☒ DHCP	☐ Static	DHCP request successful.
IPv4 Address	192.168.3.6	
Subnet Mask	255.255.255.0	
Default Gateway	192.168.3.1	
DNS Server	192.168.90.4	

PCVlan3

4.2 Test et configuration HTTP/DNS

DNS Service

☒ On ☐ Off

Resource Records

Name

Type

ARecord

Address

Add

Save

Remove

No.	Name	Type	Detail
0	neosedurh	ARecord	192.168.90.5

1.

ConfigDNS/RH

DNS

DNS Service

☒ On ☐ Off

Resource Records

Name

Type

ARecord

Address

Add

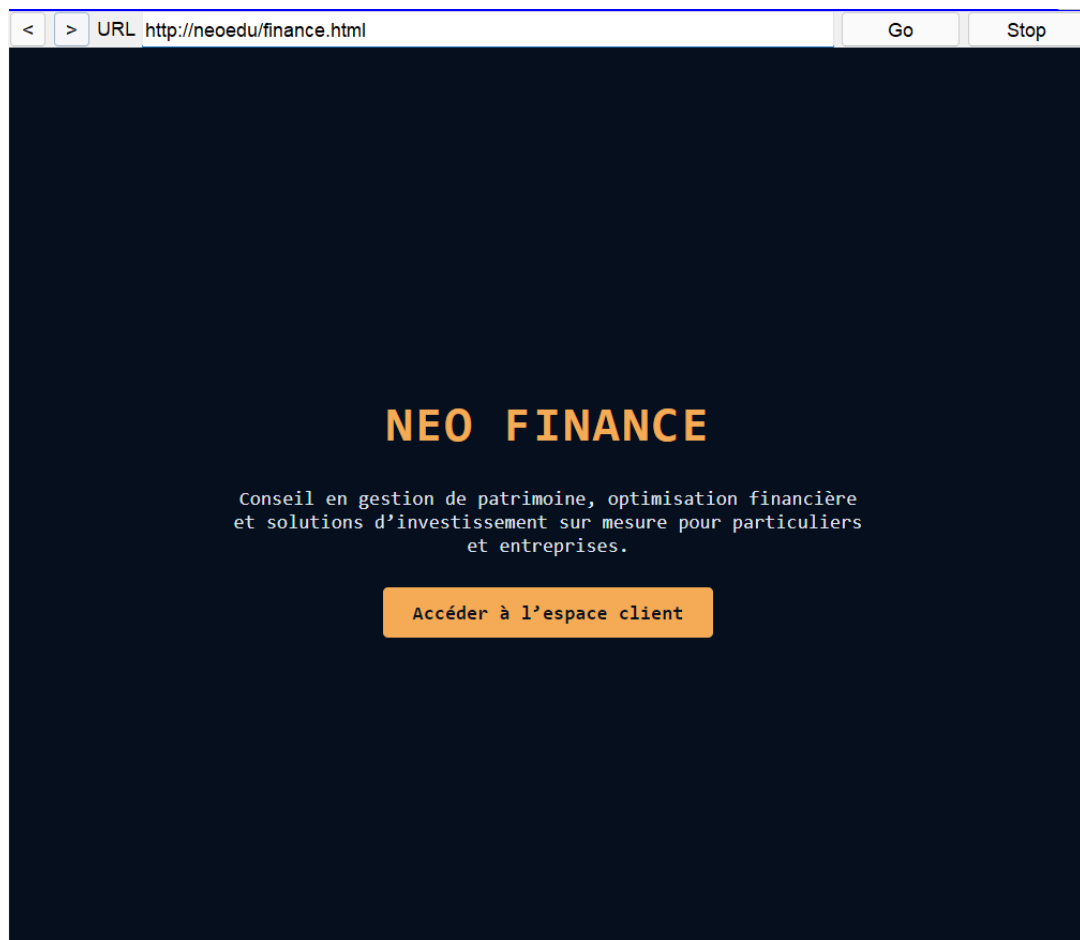
Save

Remove

No.	Name	Type	Detail
0	neoedu	ARecord	192.168.90.4

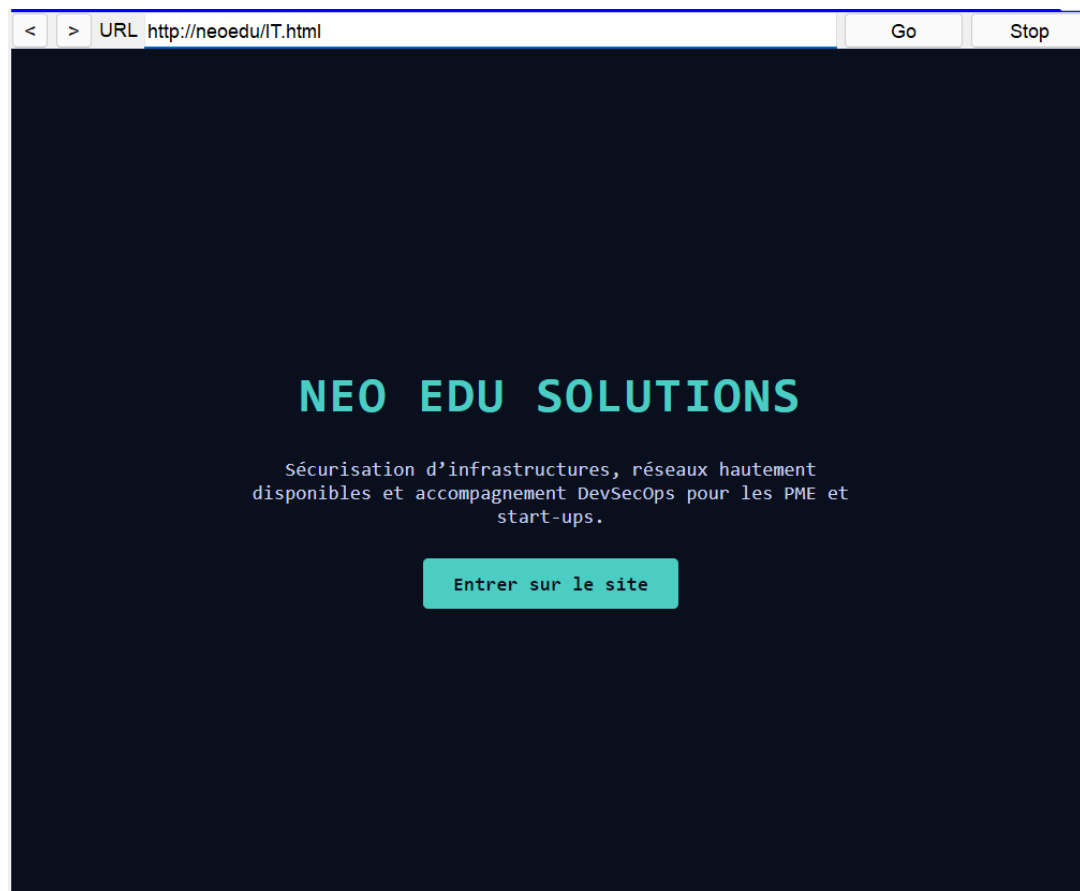
2.

ConfigDNS



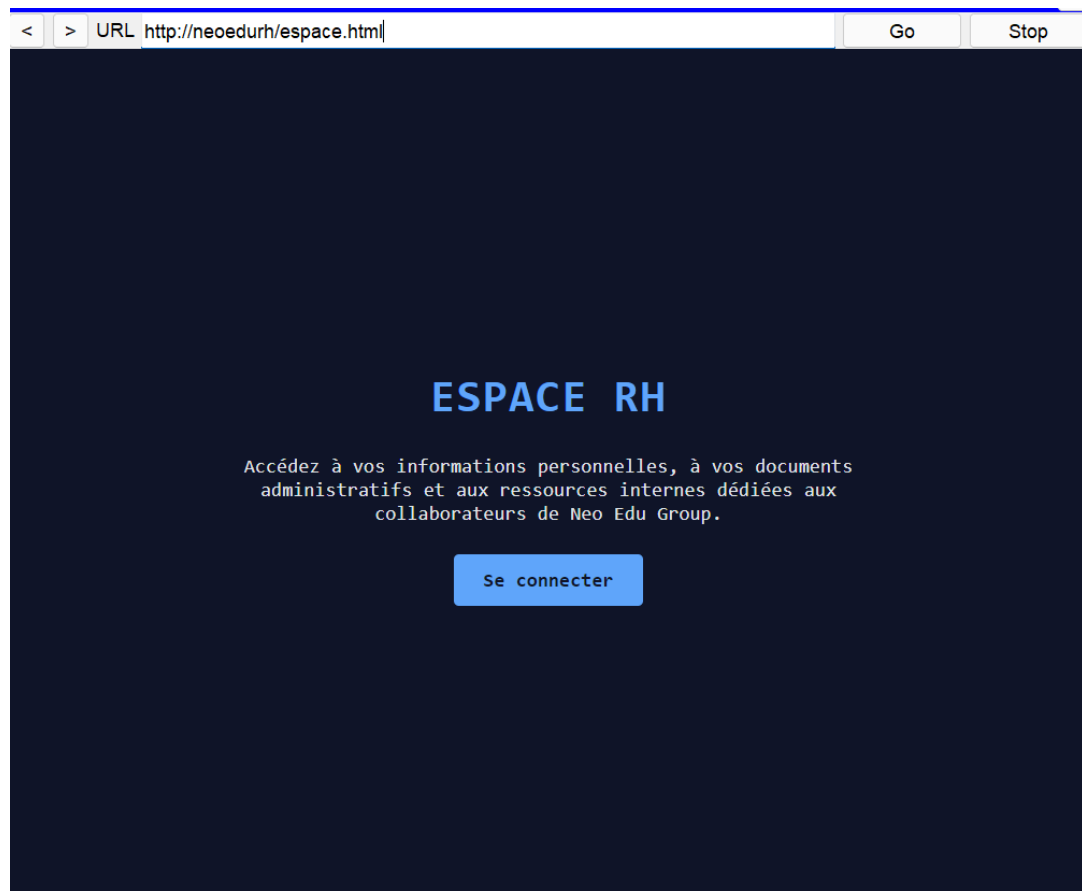
3.

Page Finance



4.

Page IT



5.

Page RH

4.3 Test TFTP

```
router1wan-running-config
router2wan-running-config
```

Enregistrement de la configuration des router

4.4 Test SYSLOG et Standby

	Time	HostName	Message
1	03.01.1993 12:00:35.584 AM	192.168.80.3	%HSRP-6-...
2	03.01.1993 12:00:37.027 AM	10.0.0.13	%HSRP-6-...
3	03.01.1993 12:00:38.829 AM	192.168.80.3	%HSRP-6-...
4	03.01.1993 12:00:39.530 AM	192.168.80.3	%HSRP-6-...
5	03.01.1993 12:00:40.106 AM	10.0.0.13	%HSRP-6-...
6	03.01.1993 12:00:40.588 AM	10.0.0.13	%HSRP-6-...
7	03.01.1993 12:00:47.472 AM	192.168.80.3	%HSRP-6-...
8	03.01.1993 12:00:51.752 AM	10.0.0.13	%HSRP-6-...
9	03.01.1993 12:01:02.425 AM	10.0.0.13	%HSRP-6-...
10	03.01.1993 12:01:14.560 AM	10.0.0.13	%HSRP-6-...
11	03.01.1993 12:01:14.562 AM	10.0.0.13	%HSRP-6-...
12	03.01.1993 12:01:44.861 AM	10.0.0.13	%HSRP-6-...
13	03.01.1993 12:01:44.863 AM	10.0.0.13	%HSRP-6-...
14	03.01.1993 12:02:23.318 AM	10.0.0.13	%HSRP-6-...

Test des syslogs servers

9	03.01.1993 12:23:30.040 AM	192.168.80.3	Grp 2 state Standby -> Active
---	----------------------------	--------------	-------------------------------

Test des standby router

4.5 Test PING entre VLAN

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Successful	Lapt...	RH2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	SN...	Router0(2)	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PCF...	PCIT	ICMP		0.000	N	2	(edit)	(delete)
	Successful	HTT...	PCFinance	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PCIT	HTTP/DN...	ICMP		0.000	N	4	(edit)	(delete)
	Successful	DH...	PCFinance	ICMP		0.000	N	5	(edit)	(delete)
	Successful	DH...	PCIT	ICMP		0.000	N	6	(edit)	(delete)
	Successful	DH...	RH2	ICMP		0.000	N	7	(edit)	(delete)
	Successful	DH...	Router0(2)	ICMP		0.000	N	8	(edit)	(delete)
	Successful	HTT...	RH2	ICMP		0.000	N	9	(edit)	(delete)

TESTPING